A Digit-Sum Congruence for Products of Twin Primes

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Abstract

We prove the congruence conjecture in OEIS A136050: the decimal digit sum of the product of any twin-prime pair other than $(3, 5)$ is congruent to eight modulo nine. The proof is elementary and applies to every such pair; it makes no assertion about the infinitude of twin primes.

Let $s(N)$ denote the decimal digit sum of a positive integer N. OEIS A136050 lists $s(p(p+2))$ over the twin-prime pairs in increasing order, beginning with $s(3 \cdot 5) = 6$. Its comment dated December 2, 2016 conjectures that each subsequent term is one less than a multiple of nine.

Theorem 1. *If $p, p+2$ are prime and $p > 3$, then*

$$s(p(p+2)) \equiv 8 \pmod 9.$$

In particular, every term of A136050 except its initial term belongs to $\{8, 17, 26, 35, \ldots\}$.

Proof. Both primes are odd. Among $p, p+1, p+2$ one number is divisible by three, and it cannot be either prime since $p > 3$. Thus $p+1$ is divisible by both two and three, giving $p = 6m - 1$ for an integer $m \geq 1$. Consequently

$$p(p+2) = (6m-1)(6m+1) = 36m^2 - 1 \equiv -1 \pmod 9.$$

For every decimal expansion $N = \sum_j d_j 10^j$,

$$N \equiv \sum_j d_j = s(N) \pmod 9,$$

which proves the claim. Since the positive digit sum is eight modulo nine, its least possible value is eight. □

The exceptional pair $(3, 5)$ gives 6, exactly the stated first term. This settles the entire recorded congruence question. It does not imply that every value $8 + 9j$ occurs, nor does it assume or prove that there are infinitely many twin primes.

The accompanying `verify.py` compares all 5000 official b-file terms with a prime-sieve computation of the twin pairs and with two independent digit-sum implementations. A trial-division prime enumeration independently checks all 342 twin pairs through 20000. Every one of the 14870 noninitial twin pairs through 2000000 satisfies the theorem. These finite audits check the implementation; the argument above holds for every pair.

References

[1] The OEIS Foundation Inc., A136050 and its 5000-term b-file, accessed October 26, 2026.